

0.1 Minimal Polynomial with Diagonalizable

Lemma 0.1.1. Let V be a finite-dimensional vector space over F , and let $T : V \rightarrow V$ be a linear operator. Suppose that the minimal polynomial p of T splits over F as:

$$p = (x - c_1)^{r_1} \cdots (x - c_k)^{r_k}, \quad (c_i \in F)$$

Then, for any proper T -invariant subspace $W < V$, there exists a vector $\alpha \in V$ such that $\alpha \notin W$ and $(T - c_i I)(\alpha) \in W$ for some eigenvalue c_i .

Proof. Let $\beta \in V \setminus W$ be given, and let $g \in F[x]$ be the T -conductor of β into W (i.e., the monic polynomial of least degree such that $g(T)\beta \in W$). Since g divides the minimal polynomial p and $\beta \notin W$, g is not constant. Therefore, g must be of the form:

$$g = (x - c_1)^{e_1} \cdots (x - c_k)^{e_k}, \quad (0 \leq e_i \leq r_i)$$

where at least one e_i is positive. Pick j such that $e_j > 0$. Then we can write $g = (x - c_j) \cdot h$ where $h = g/(x - c_j) \in F[x]$. By the definition of the T -conductor, $h(T)\beta \notin W$. Let $\alpha = h(T)\beta$. Then:

$$(T - c_j I)\alpha = (T - c_j I)h(T)\beta = g(T)\beta \in W.$$

Thus, α is the required vector. □

Theorem 0.1.2. Let V be a finite-dimensional vector space over F , and $T : V \rightarrow V$ be a linear operator. T is triangulable if and only if the minimal polynomial of T splits over F .

Proof. ($\Rightarrow$) If T is triangulable with respect to a basis β , then the matrix $[T]_\beta$ is upper triangular. The characteristic polynomial is $f(x) = \det([T]_\beta - xI) = \prod_{i=1}^n (A_{ii} - x)$, which splits over F . Since the minimal polynomial divides the characteristic polynomial, it also splits.

($\Leftarrow$) Conversely, choose $\alpha_1 \in V$ such that $\alpha_1 \neq 0$ and $(T - c_1 I)\alpha_1 = 0$ for some eigenvalue $c_1 \in F$. Let $W_1 = \langle \alpha_1 \rangle$, which is T -invariant. By the Lemma, there exists $\alpha_2 \in V \setminus W_1$ such that $(T - c_2 I)\alpha_2 \in W_1$. Then $W_2 = \langle \alpha_1, \alpha_2 \rangle$ is T -invariant. Inductively, we obtain a basis $\{\alpha_1, \dots, \alpha_n\}$ such that:

$$\begin{aligned} T(\alpha_1) &= c_1 \alpha_1 \\ T(\alpha_2) &= a_{12} \alpha_1 + c_2 \alpha_2 \\ &\vdots \\ T(\alpha_n) &= a_{1n} \alpha_1 + a_{2n} \alpha_2 + \cdots + c_n \alpha_n. \end{aligned}$$

In this basis, $[T]_\beta$ is upper triangular. □

Theorem 0.1.3. Let V be a finite-dimensional vector space over F , and $T : V \rightarrow V$ be a linear operator. T is diagonalizable if and only if the minimal polynomial of T splits and is separable (i.e., has no multiple roots).

Proof. ($\Rightarrow$) Suppose T is diagonalizable. Let $c_1, \dots, c_k$ be distinct eigenvalues. For any eigenvector α , $(T - c_i I)\alpha = 0$ for some i . Let $p = (x - c_1) \cdots (x - c_k)$. Then $p(T)\alpha = \prod (T - c_j I)\alpha = 0$ because the operators $(T - c_j I)$ commute. Since any vector $v \in V$ is a linear combination of eigenvectors, $p(T)v = 0$ for all v . Thus $p(T) = 0$, and the minimal polynomial, which must have these c_i as roots and divide p , has no multiple roots.

($\Leftarrow$) Conversely, let W be the subspace spanned by all eigenvectors of T . Suppose $W \neq V$. By construction, W is T -invariant. By the Lemma, there exists $\alpha \in V \setminus W$ and an eigenvalue c_j such that $\beta = (T - c_j I)\alpha \in W$. Since $\beta \in W$, we can write $\beta = \beta_1 + \cdots + \beta_k$ where $T(\beta_i) = c_i \beta_i$. For any polynomial $h \in F[x]$, $h(T)\beta = \sum h(c_i)\beta_i$.

Let the minimal polynomial be $p = (x - c_j)q$. Also, since $q(c_j) \neq 0$ (no multiple roots), we can write $q - q(c_j) = (x - c_j)h$ for some $h \in F[x]$. Then:

$$q(T)\alpha - q(c_j)\alpha = h(T)(T - c_j I)\alpha = h(T)\beta \in W.$$

However, $0 = p(T)\alpha = (T - c_j I)q(T)\alpha$. This means $q(T)\alpha$ is an eigenvector for c_j , so $q(T)\alpha \in W$. Since $q(T)\alpha \in W$ and $h(T)\beta \in W$, we must have $q(c_j)\alpha \in W$. Since $q(c_j) \neq 0$, this implies $\alpha \in W$, which contradicts $\alpha \notin W$. Thus $W = V$, and T is diagonalizable. $\square$